

MANIK GOYAL

Backend Software Engineer | .NET, C#, Azure, Angular

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2 years of production experience building **.NET 8 services, REST APIs and Angular interfaces** for a video analytics platform running across **400+ cameras and 200+ servers**. Works daily in C#, message-based architectures and PostgreSQL, scaling sustained event throughput **16x** and cutting analytics export runtime by **92%**.

TECHNICAL SKILLS

Languages: C# (C Sharp), C++17, SQL, TypeScript, JavaScript, HTML5, CSS3

Backend & Frameworks: .NET 8/7, ASP.NET Core, Entity Framework Core, LINQ, SignalR, REST APIs, Microservices, Distributed Systems

Frontend: Angular 18, Angular Material, RxJS, TypeScript, NPM package authoring, Component Libraries

Messaging & Streaming: RabbitMQ, MQTT, Pub/Sub, Asynchronous Messaging, Dead-Letter Queues, SignalR Hubs, WebRTC, FFmpeg

Databases: PostgreSQL, TimescaleDB, SQL Query Optimization, Indexing, Partitioning, Compression, Milvus (vector search)

Cloud & DevOps: Microsoft Azure, Docker, GitHub Actions, CI/CD, Linux, Grafana, Prometheus, Observability

Practices: System Design, Clean Architecture, SOLID, Design Patterns, API Design, Performance Tuning, Code Review, Agile/Scrum, Unit Testing

Spoken: English (Professional), Hindi (Native), German (Basic)

PROFESSIONAL EXPERIENCE

Software Engineer, I2V Systems Pvt. Ltd.

Oct 2024 - Present

Gurugram, India | .NET, C#, RabbitMQ, PostgreSQL, Angular, Docker

- Led development of the Analytics Manager service in **.NET** for a **400+ camera** video analytics platform, coordinating ANPR, face recognition and intrusion events across **8+** backend services.
- Diagnosed message-processing bottlenecks and redesigned broker consumption around event-specific queues with dead-letter handling, lifting sustained throughput from **30 to 500+ events/sec (16x)** while isolating failed messages for replay.
- Architected reusable messaging and storage abstraction libraries spanning **RabbitMQ, MQTT, local disk, NAS and cloud object storage**, adopted by **4+ services** and cutting new-service integration from days to hours.
- Optimized PostgreSQL and TimescaleDB ingestion with hypertables, native compression, retention policies and composite indexes, raising camera-frame processing from **70 to 300+ frames/sec**; tuned analytics database indexes to cut query latency by **35%**.
- Implemented Milvus vector similarity search for face enrollment, watchlists and deduplication, reducing vector-index memory consumption by **75%** while preserving match accuracy.
- Reengineered analytics exports over **100K+ records** with streaming reads and incremental batch processing, cutting execution time from **12 minutes to under 1 minute (92% faster)** and eliminating out-of-memory failures.
- Built bulk enrollment workflows for **5,000+ records** with server-side validation and live **SignalR** progress streaming; integrated government e-challan REST APIs into the ANPR violation pipeline.
- Published **10+ reusable Angular/NPM component libraries** consumed by web apps across a **200+ server** deployment serving **60+ concurrent users**, standardizing UI behavior across products.
- Automated Docker-based release and changelog workflows with GitHub Actions, reducing release turnaround by **40%**, and instrumented production services with Grafana and Prometheus dashboards and alerts.
- Mentored **2** junior engineers on Clean Architecture and asynchronous C# patterns, and drove code reviews that cut QA-reported regressions.

PROJECTS

The Streamer, Low-Latency Video Streaming Platform

2025

C++17, FFmpeg, WebRTC, .NET 7, Docker, GitHub Actions

- Developed a streaming platform delivering **sub-200 ms** camera playback, decoding and remuxing H.264 video with C++17 and FFmpeg for both live and recorded sources.
- Migrated transport from WebSocket streaming to **WebRTC/RTP** using libdatachannel, adding HTTP signaling and data-channel seek, pause and playback-speed controls.
- Standardized builds and releases with CMake, vcpkg, Docker and GitHub Actions, including Windows MSI packaging for one-command deployment.

ParkNest, Peer-to-Peer Parking Marketplace

2025

.NET 8, ASP.NET Core, PostgreSQL, Angular, Asynchronous Messaging

- Engineered a two-sided marketplace on **ASP.NET Core** supporting host listings, renter bookings and prepaid credit settlement over a normalized PostgreSQL schema.
- Designed optimistic-concurrency booking safeguards and an idempotent credit ledger that prevent double-booking and duplicate transactions under overlapping requests.

EDUCATION

Bachelor of Technology (B.Tech), Information Technology

Dec 2021 - Jun 2025

University School of Information, Communication and Technology (USICT), GGSIP University, Delhi | **CGPA: 8.55/10**

CERTIFICATIONS & ACHIEVEMENTS

Postman API Fundamentals Certification - API design, documentation and automated testing workflows.

LeetCode: solved **800+** data structures and algorithms problems across arrays, strings, graphs, trees and algorithm design.

Adobe GenSolve: placed in the **top 5 percentile** nationally in the algorithmic coding competition.